Diffusion: Realisation, Analysis and Evaluation

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Introduction

This paper attempts to deal with various aspects of the theory and practice of music diffusion. Last year's conference was centred on space and I want to focus on the interfaces by which we achieve spatialisation and the interface between spatialised sonic material and our perception and understanding. Generally speaking diffusion is the process whereby the inherent spatial properties in a piece of audio may be realised through electro-acoustic means during concert presentation. It may also be a process whereby spatial properties are imposed on material where no strong spatiality is present. The term itself implies a particular approach characterised by a diffuse/ de-correlated sound-field. I would like to argue that while this approach has many benefits, and probably results from an understanding of the practical limitations of physical electro-acoustics, it is by no means the only answer. What follows is a wide ranging discussion of the various aspects informing the practice, analysis and evaluation of electro-acoustic diffusion.

Realisation

I will start with the practical aspects, the realisation of concert diffusion. Last years conference dealt with this issue in some detail. I would like to bring together some ideas from my experience. Firstly, I think it's important to have some understanding of the history of electroacoustic spatialisation and more broadly the history of the use of space in musical practice.

The very nature of the acoustic transmission of sound within air as a medium invites composers to manipulate spatial properties in performance. As soon as more than one performer is present there is a spatial element. A quick fire survey of the Western canon would identify antiphon in medieval church music particularly Gregorian chant; followed on with more elaborate applications of antiphonal composition in the Baroque and classical periods; St. Mark's in Venice with its spatially opposed organs was the site of some development in the late Renaissance / early Baroque, of the antiphonal devices especially associated with the Gabrielis, Andrea and his nephew Giovanni. They pioneered a polyphonic style where the various melodic lines were presented by spatially distinct choral and instrumental groupings. Other famous examples of the use of spatial effects include the antiphonal choral effects in J.S.Bach's St. Matthew Passion and Mozart's Serenade in D for 4 Orchestras with its motivic interplay between spatially separated instrumental groups. The inevitable spatial effects of Mahler's Symphony No.8 in E flat 'Symphony of a Thousand' are really more a result of the sheer expanse of performers than an effect by design. Finally, the spatial location of musical sources has often been exploited in the theatre. There are many examples of on or off stage bands in opera. All this has led to an expectation of spatial properties in musical performance.

With the advent of electro-acoustic devices for the recording, transmission and re-play of musical material, it is not surprising that a more extensive presentation of space was

expected by the public than that which could be encoded by variations in level, timbre and reverberation in a mono signal. It took sixty years for the music publishing industry to release stereophonic recordings although the theory of stereo spatial encoding had been around for at least twenty years prior. Popular music performers were quick to adopt recordings as the site for their compositional process with strange and clumsy use of stereo exemplified by such works as The Beatles "Sgt Pepper's" album.

Of course, I have jumped over the infiltration of electroacoustic devices in to the music academy. The potentially liberating invention of the loudspeaker held a unique fascination for composers inclined toward modernism. It was not until after the second world war, of course, in the spirit of redevelopment and turning away from the past, that the use of loudspeakers in musical composition became more common. The use of the tape recorder as an adjunct to acoustic instruments or as an alternative, as in the work of Pierre Schaffer, became more common. However, electroacoustic composition with space as an integral musical component is a distinct form. Another quick fire list of the more famous examples includes (excuse my pronunciation): Stockhausen's "Gesang der Jünglinge" of 1956 which includes five loudspeaker locations, Kontakte 1960 includes a four track tape. The Philips pavilion for the 1958 Brussels world fair, designed by Iannis Xenakis and Le Corbusier included an 11 channel 425 loudspeaker sound system for which Edgar Vàrese composed Poème Electronique and Xenakis produced Concret PH. Stockhausen produced various pieces for the German pavilion at the Osaka EXPO 70. The pavilion was a geodesic dome with loudspeakers at nearly every vertex and integral sound control equipment. EXPO 70 also included the 800 speaker installation in the Japanese Steel pavilion. This system was used by Xenakis for his 12 channel composition Hibiki Hana Ma. The idea of introducing the spatial complexity of an orchestra into electroacoustic performance has been explored by the Groupe de Musique Expèrimentale de Bourges with their multi-loudspeaker Gmebaphone. A similar idea is seen in the Acousmonium of the Groupe de Recherches Musicales. Pierre Boulez utilised a multi-speaker system suspended in the auditorium for his Rèpons. In the States, John Chowning developed software to control spatial location and panning in a four speaker system.

Of course, I've left out lots of vital examples but over this period a range of practice evolved. At the same time in the more commercial field of music publishing, various methods were being employed to improve the fidelity of recordings. Engineers attempted to provide more and more accurate mapping of auditorium acoustics into home reproduction.

Composers such as Stockhausen, desperate to specify every aspect of the musical performance also did battle with the vagaries of auditorium acoustics. Repeatable presentation of spatial encodings continue to defy practitioners in all fields. Michael Gerzon's brilliant meta-theory of spatial encoding which supports the practice of ambisonic recording provides a perfect framework for spatial specification. However, it fails completely to overcome the effects of auditorium acoustics for replay and let's face it, for those who have done experiments with ambisonics, you can never have too many speakers or too little reflection in your performance space. These technologies, however, have become more subtle and refined and proven to very effective when employed carefully.

However, many composers and sound designers working with space at a musical level want more than subtlety. They want bold gestures, they need as much support

for their fragile musical intentions as an electro-acoustic system can provide. Hence we find what appear to the acousticians as fairly crude multi-channel systems such as Dolby surround and those used in much of today's acousmatic and electro-acoustic performance.

Accurate spatial reproduction requires highly correlated, phase accurate, directivity controlled multi-channel speaker systems. Theatre sound designers of the past two decades have been working away at the margins, employing psycho-acoustic principals such as the Haas effect and utilising more and more sophisticated measurement systems, to trick audiences into believing the realism or transparency of their sound reinforcement systems.

Parallel research into psycho-acoustics has shown that our ability to understand and interpret musical information in polyphonic writing is aided by spatial de-correlation. Spatially separated steams can be processed in parallel retaining more of the discrete informational content of the individual streams than those collapsed into one spatially coherent stream.

Essentially, there are three approaches to spatial manipulation: Firstly, we can build a soundfield from a number of discrete, de-correlated, spatially separated streams or sources. This is what happens with an instrumental ensemble on stage. The acoustic interaction of the auditorium, however, produces diffusion, that is, a build up of discrete reflections which increases the correlation of the acoustic information coming from one location in space, and therefore fusing the instrument streams, blending timbres. The acoustics of an auditorium given by its overall volume and the absorptivity of its surfaces and their relative orientation thus producing differing reverberation times, ratios of direct to reflected sound and onset, density or diffusion of reflections, effects the correlation of the discrete streams and therefore the perceived spatialisation.

A second approach to spatialisation takes a single coherent stream and presents it from a number of spatially separated acoustic sources, here I'm talking specifically about loudspeakers. This might seem like a totally artificial type of spatialisation that is unlikely to occur in nature. But, I think in terms of our perception, it probably comes close to certain modes of listening. Our brains ability to perceive separate auditory streams is determined both by their number and spectro-morphology, and the type of attention that is paid to them. Later, I will introduce the concept of *modes of listening* or *listening relationships* which attempt to describe the broad overlapping categories of attention paid to musical material. In certain circumstances streams are fused in our perception, if these fused streams originate from separate acoustic sources we may tend to feel surrounded by a particular sound. Other instances of this type of perception are generated in highly diffuse sound fields and by the psycho-acoustic perception of low-frequency sounds.

The third type of spatialisation involves moving sources. The illusion of a moving source is usually achieve by proportional signal panning of a single stream across a number of spatially separated loudspeakers.

Each of these types of spatialisation is fairly distinct in terms of its coding or imaginative associations.

I have mentioned the term correlation which will be familiar to the mathematicians and engineers in the audience but probably needs a bit of conceptual unpacking. Broadly speaking, statistical correlation describes a continuous measure of matched groupings in a number of data sets. In signal processing in both discrete and continuous time systems correlation is a measure of the average coincidence of phase and amplitude

values of two signals. It is the continuous average of the products of two or more signals. Correlation is used to describe a stereo signal to gauge its stereo spatial content. In this case, a correlation value of one indicates a mono signal. Minus one is 180 degrees out of phase, and a zero correlation indicates highly de-correlated material. A reasonably high positive values is usual in stereo music recordings.

Correlation is a useful concept to keep in mind when discussing music diffusion. Consider two examples: first a single audio stream is sent to eight loudspeakers surrounding the audience, the stream contains a solo violin. This is an artificial sound field, flat, strange, maybe dreamlike but highly spatially correlated. The second example uses eight separate streams, mono recordings of cicadas. Each stream is sent to one of eight loudspeakers. Here the soundfield is alive (in more ways than one), highly de-correlated, highly textured exciting and natural. Of course, there is a range of possibilities between these two extremes.

As I mentioned the diffusion of a sound field is directly related to the interaction of the electro-acoustic devices – the loudspeakers – and the architecture. This in turn influences the level of correlation that may be achieved in a given auditorium. Frequency dependant absorption and specular reflection of architectural materials inter-react with the frequency dependant directivity or beamwidth of loudspeakers. The unpredictability of these relationships has led the designers of diffusion systems to make a range of idiosyncratic choices in the selection and deployment of electroacoustic devices. A classic example being the Acousmonium of the GRM.

The choice of playback format and performance interface is further influenced by musicological factors, including distinctions between performance and composition, fixed/notated and improvised performance processes

The dichotomy between a compositional approach which begins from the intrinsic sonic properties of a sound object and develops through experimentation towards larger forms, and an opposite approach which begins from formal objectives and then moves to the sound material with which these forms might be realized, was identified by Pierre Schaeffer in his early researches into musique concrète. This dichotomy is characteristic of the divergent approaches to the composition and performance of electronic and electroacoustic music that have existed throughout its fifty year history. The culturally established norms of notated composition and interpretive performance are still influential in discussions of contemporary practice. Similarly, the distinction between composition and improvisation are still held in some quarters despite the blurring of these boundaries that has occurred through practice in the electronic music studio.

One interesting aspect of this is the traditional location of formal and constructional properties of a composition in the score. This represents the work of the *composer*. Conversely, in traditional performance practice the expressive nuance or gestural shaping of the sonic events that comprise the music are in the domain of the *performer*. Electronic and tape composition fuses these aspects of the music in the work of the composer in the studio. The composer is concerned both with the large scale formal aspects of the composition and the discrete gestural shaping of sonic events. The studio environment allows the composer to experiment, evaluate and modify each element of the musical material. With the appropriate direct manipulation control device the studio composer can *perform* elements of the composition and record the results.

This situation radically changes the traditional concept of the role of the composer in two related and fundamental ways. Firstly, the composer is working directly with the materials of his composition. The composer is manipulating the sonic material rather than merely encoding directions for its production in the score. Secondly, and as a result of the first point, the composer is *performing* the composition in the studio through a process of experimentation and evaluation. This results in a blurring of the distinctions between composition, performance and improvisation.

The unpredictability of spatial outcomes in different auditoria encourages the composer to adapt the spatialisation of a piece for each new performance. The lack of standardisation of replay systems also makes this a necessity. The ease of production and dissemination for two channel or stereo media dictates this choice as well. Behind all of this though, there is the need for reaffirmation of the composer as performer and the perceived need for theatricality in the concert space. The performed diffusion as has become common in the UK and Europe is the result of all these factors.

There are of course, other alternatives. Real-time digital signal processing is giving us the means to record and edit multi-channel diffusions with greater ease. Replay and mixing systems such as LCS, AudioBox and Protools allows for rapid experimentation and modification of existing diffusions. AudioBox control software allows for automatic translation of a diffusion from one physical speaker arrangement to another. This sort of processing brings the studio composition/performance practice of the composer into the concert hall. The diffusion and indeed the mix is no longer fixed on tape and the composers intentions may be more fully realised through rehearsal and refinement in the actual performance space. Of course these developments do not negate the validity of performed diffusion and the altered acoustics of a full auditorium will always encourage the corrective participation of virtuosic diffusion performance.

Analysis

The analysis of music diffusion or soundstage presentation in commercial music recordings has a long history and well established principals. The theoretical basis for analysis of diffusion in electro-acoustic and acousmatic music is also well established, however, its practice and the drawing together of the theoretical strands which must underlie its musicology are in their infancy. The topic has been developed by authors such as Denis Smalley and Trevor Wishart and Simon Emmerson but they have avoided some of the practical considerations outlined above. The field of the psychology of auditory scene analysis and the application of its insights to the development and presentation of scene based structures in musical composition is another area ripe for development. The principals of soundscape design and analysis should influence judgments on musical content and the intrinsic and extrinsic relations and structures inherent in its elements. The inter-relations between scene analysis and scene description in the soundscape should further be exposed when informed by an analysis of the spectro-morphological properties of the sonic materials. This approach should reveal aspects of the source and genesis of sound objects and more complex sound structures and their spatial inter-relations. Spectro-morphological analysis underpins the discussion of spatio-morphology (to borrow Smalley's jargon) and provides further theoretical basis for practical analysis. Finally, a poetics of audible space and an image analysis of spatial relations should

reveal a rich poetic source of resonance and identification. Image, like correlation, is a densely packed concept. It brings with it notions of spatial perception, literary analysis, and the shimmering inversions of subject and object which make the greatest examples of electronic music such rewarding listening.
Spatial motion as gesture is an obvious starting point. And an analysis of levels of gestural surrogacy, as Denis Smalley describes, provides a frame work. In Smalley's spectro-morphology low order gestural surrogates provide a direct audible link between the causal mechanism and its sounding outcome. As we move higher in the order of gestural surrogacy, possibly through the application of some signal processing treatment changing the amplitude envelope of a sound, we effectively disguise the causal source. So it is with spatial motion. The scale of the scene is determined by the possible spatial motion of the sources. The spectro-morphology must also relate to the spatial motion. A fly must be in motion otherwise it is silent.

In the wide-open sound world of electroacoustic music we need structured methodologies to facilitate analysis and I think a return to the philosophical origins of musique concrete may help to show the way. The phenomenology of Henri Bergson influenced writers such as Gaston Bachelard whose psychoanalytical approach to material imagination must surely have prompted Schaeffer in his typology of spectral forms. An attempt to measure the imaginative charge of spatial gesture in electroacoustic art would do well to start from Bachelard's "Poetics".

Following Schaeffer's analysis of *modes of listening*, Smalley approaches the analysis of extrinsic referents in terms of the listeners perception. Smalley identifies types of relationship that the listener may have with the musical material (Smalley 1996). These include the *indicative relationship* in which the sound carries information or a message for the listener. The listener may enter into this relationship actively or passively depending on whether the sound event is expected or not. The second relationship is the *reflexive relationship*. This is a fairly one-sided relationship in which the listener is mainly aware of the emotional or affective impact that the sound has on him. The final relationship described by Smalley is the *interactive relationship*. In this situation the listener is actively engaged in exploring the sound and its intrinsic properties and relationships to other sounds and events. According to Smalley most listening is characterised by the first two relationships. The interactive relationship is characteristic of the type of listening engaged in during a concert or other concentrated listening. During normal listening the relationship will shift between these three depending on the sound material and the conscious attention of the listener.

It is the scope for development of the subtle use of the indicative relationship that Smalley spends some time developing. He introduces nine classes within what he terms as the *indicative field* or *indicative networks*. These classifications include, in Smalley's terminology, *gesture, utterance, behaviour, energy, motion, object/substance, environment, vision, and space.*

The interplay of these listening relationships is what spatialisation does best. The lack of strong coding of spatial properties, should not and does not deter the composer or the critic from a thorough exploration of these principals.

Evaluation

The final area I would like to discuss suggests scope for the evaluation of diffusion as a musical property.

The analytical principals I have introduced should begin to help the critic to navigate in the unmapped territory of the wide-open-soundworld. Acousmatic music presents the paradox of a humanist tradition realised by material means. The materializing and abstracting force of technology must be harnessed and directed by the imaginative composer. The corruption of our listening culture can be exemplified by our acceptance of machine originated cyclic panning motions, made acceptable by our constant exposure to these patterns of motion in both commercial music and commercial machines.

Diffusion practice must start from the sonic material, from the perception of sonic phenomena. Just as the pattern forming processes of our minds revert to sounding bodies and gestural causes. In our imaginative life and in the world of dreams, spaces undergo endless transformations. These transformations parallel the poetic, mythic and psychic development represented by a rich and powerful acousmatic art.

Spatial expression that is too divorced from gestural sources will appear cold and sterile. These properties have often been exploited or exposed in the machine dominate culture of computer music and they have their place in the expression of these principals.

The successful critic will be aware of the physical limitations of performance and presentation systems at the level of the physical interface. The critic will also be concerned with the imaginative interface between the composers intentions and the listeners perceptions and subjective spatial images.

The need to develop and discuss the principles of diffusion, its realisation, analysis and evaluation have provided the motivation for this presentation and I thank you all for your attention.

References

Bachelard, G. 1994. *The Poetics of Space* Beacon Press, Boston USA.

Chion, M. 1994. *Audio-Vision - Sound on Screen* Columbia University Press, New York USA.

Dack, J. 1994. *Pierre Schaeffer and the Significance of Radiophonic Art* Contemporary Music Review 10:2 pp. 3-11 Harwood Academic Publishers, Amsterdam Netherlands.

Dahlhaus, C. 1982. *Esthetics of Music* Cambridge University Press, Cambridge U.K.

Emmerson, S. 1996. *Local/field: Towards A Typology Of Live Electronic Music* Journal of Electroacoustic Music Vol.9 January 1996. pp.10-12, Sonic Arts Network, London UK.

Griffiths, P 1994. *Modern Music: A Consice History* Thames and Hudson, London UK.

Griffiths, P 1995. *Modern Music and After* Oxford University Press, Oxford UK.

Griffiths, P. 1986. *The Thames and Hudson Encyclopedia of 20th-Century Music* Thames and Hudson, London.

Grout, D.J. & Palisca, C. V. 1988. *A History of Western Music* Fourth Edition J.M.Dent and Sons, London.

Maconie, R. 1976. *The works of Karlheinz Stockhausen* Oxford University Press, London UK.

Manning, P 1993. *Electronic and Computer Music* Oxford University Press, Oxford UK.

Minsky, M. 1981. *Music, Mind, and Meaning* in Roads, C. ed. 1989 *The Music Machine* MIT Press, Massachusetts USA

Smalley, D. 1986. *Spectromorphology and structuring processes* in Emmerson, S. ed. *The Language of Electroacoustic Music* pp.61-93 Macmillan Press, UK.

Smalley, D. 1994. *Defining timbre - refining timbre* Contemporary Music Review 10:2 pp. 35-48 Harwood Academic Publishers, Amsterdam Netherlands.

Smalley, D. 1996. *The Listening Imagination: Listening in the Electroacoustic Era* Contemporary Music Review 13:2 Harwood Academic Publishers, Amsterdam Netherlands

Smalley, D. 1997. *Spectromorphology: explaining sound-shapes* Organised Sound 2:2 Cambridge University Press, Cambridge UK.
Truax, B. 1984. *Acoustic Communication* Ablex Publishing Corporation, Norwood New Jersey USA.
Wishart, T. 1986. *Sound symbols and landscapes* in Emmerson, S. ed. *The Language of Electroacoustic Music* pp. 41-60 Macmillan Press, UK.
Wishart, T. 1996. *On Sonic Art.* Harwood Academic Publishers, Amsterdam Netherlands.